



EXD ASSIGNMENT 2

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PROJECT OVERVIEW

The ICT AR Photo Booth Kiosk is an interactive self-service installation that allows students and visitors to take themed photos using augmented reality (AR) filters. It enhances engagement during school events such as Open House and Orientation while promoting Ngee Ann ICT's identity.

PROBLEM STATEMENT

School events often lack interactive engagement points that leave visitors with a memorable takeaway. Most photo-taking relies on personal phones, which limits branding and shared experiences.

PROPOSED SOLUTION

The AR Photo Booth Kiosk provides a fun, branded, and shareable experience using AR filters and themed frames, encouraging interaction and digital sharing.

DESIGN RATIONALE

The ICT AR Photo Booth Kiosk is designed to be simple, engaging, and easy to use for students and visitors. A clear step-by-step interface with large buttons ensures smooth interaction, even for first-time users. AR filters and ICT-themed visuals enhance engagement and encourage photo sharing, helping to promote school identity. The kiosk’s slim, modern design and ergonomic hardware placement ensure it fits well within the ICT environment while remaining accessible and practical.

USER PERSONA

ISABELLE GEORGE

PROFILE

Age

:

16

Education

:

Secondary 4 Student

Occupation

:

Prospective Student

Device Usage

:

Smartphone, Instagram, TikTok



BIOGRAPHY

Isabelle is a final-year secondary school student visiting the School of InfoComm Technology during an Open House event. She is interested in technology-related courses but feels slightly overwhelmed by the amount of information presented. She enjoys interactive experiences and often documents events through photos and social media.

MOTIVATIONS

Isabelle is motivated by interactive and visually engaging experiences. A fun AR photo booth encourages her to spend more time at the booth, share her experience online, and associate NP ICT with creativity and innovation.

GOALS

Have a fun and memorable Open House experience

Take photos to share on social media

Feel welcomed and engaged by the school

Create a positive first impression of NP ICT

FRUSTRATIONS

Long queues or confusing installations

Information overload during school events

Experiences that feel boring or outdated

Complicated interfaces that require assistance

PERSONALITY

Introvert

Extrovert

Thinking

Feeling

Judging

Perceiving

Sensing

Intuition

TECHNOLOGY

Software

Social Media

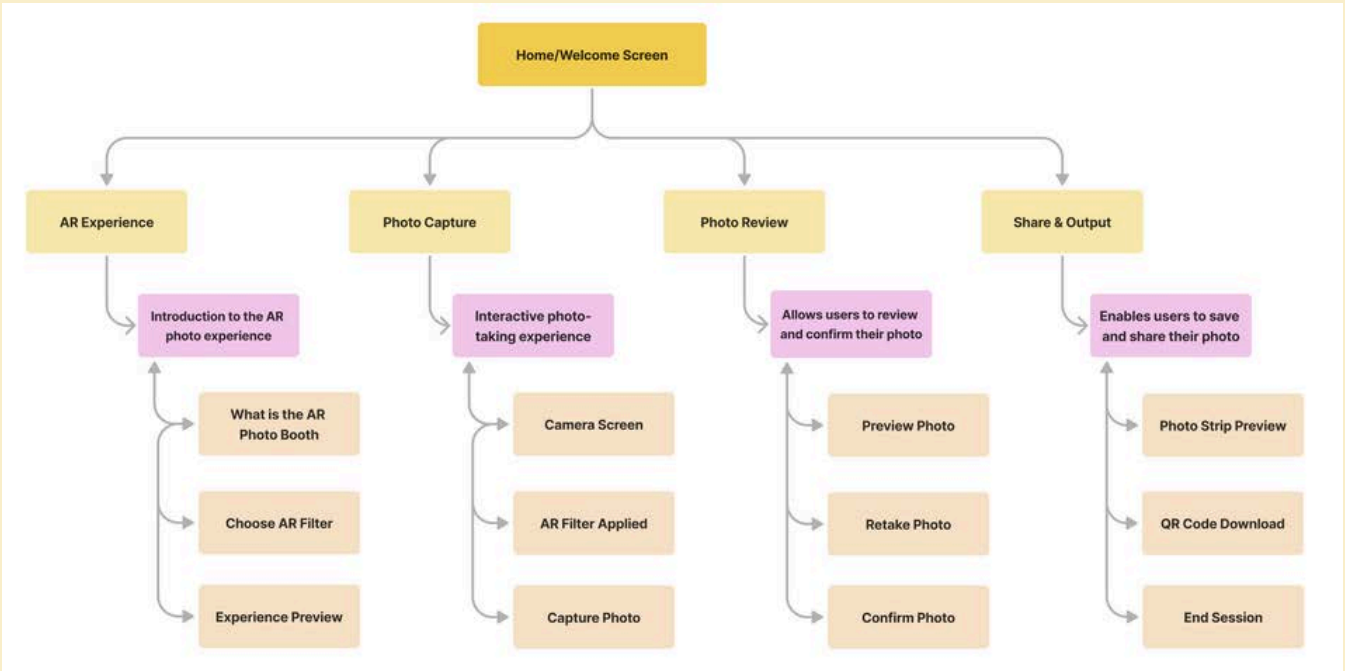
Mobile App

"I really love taking pictures! It doesn't matter if I am taking them myself or with others, I just enjoy posting and snapping all the time!"

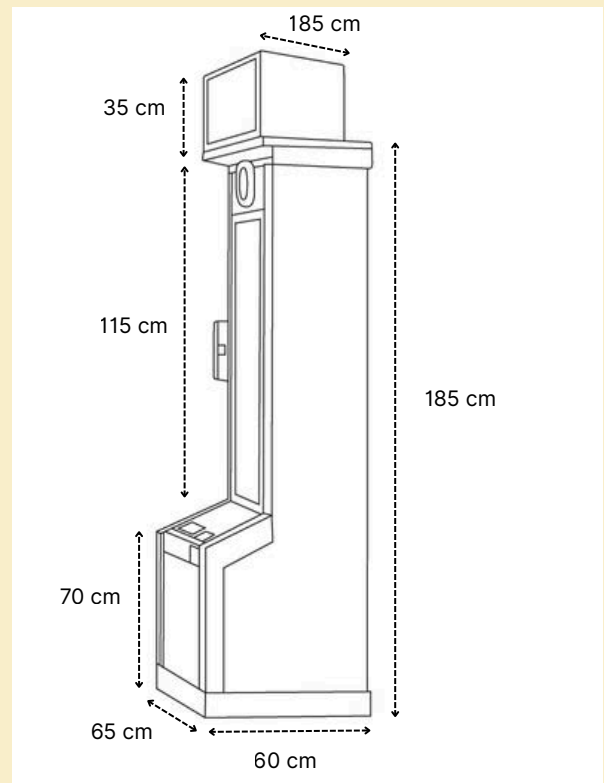
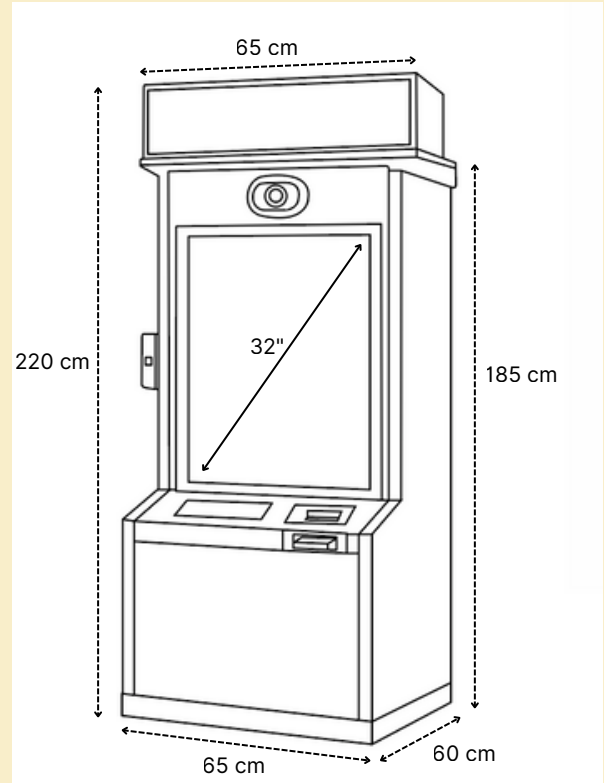
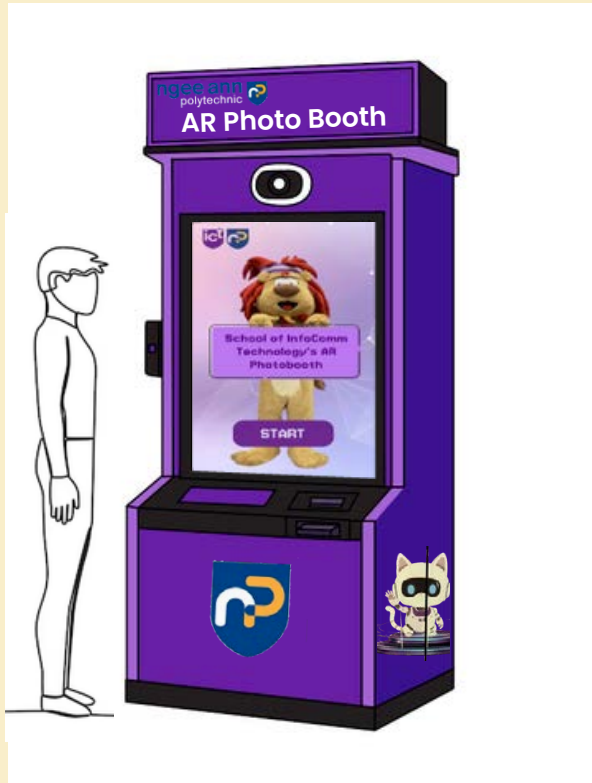
USER JOURNEY

Scope and goal	Isabelle is visiting NP ICT during Open House and notices the AR Photo Booth Kiosk. She hopes to leave with cute pictures.						
Phases	Awareness	Approach	Filter Selection	Photo Capture	Review & Confirm	Share/Download	Exit
User actions	Notices the kiosk while walking past	Walks towards the kiosk	Browses and selects an AR filter	Poses and takes photo	Reviews photo and confirms	Scans QR code or emails photo	Leaves the kiosk area
Thoughts and feelings	"This looks fun and interesting"	Curious and excited	Enjoys the visuals	Fun and engaged	Satisfied with result	Happy and accomplished	Positive impression of NP ICT
Painpoints	Unsure what the kiosk does	Afraid it might be complicated	Too many choices could be overwhelming	Camera positioning or lighting issues	Wants quick retake option	Slow sharing process	Fast QR download and clear instructions
Opportunities	Eye-catching visuals and clear signage	Simple welcome screen with clear instructions	Limit filter options and show previews	Auto-adjust camera and LED lighting	Clear retake and confirm buttons	Fast QR download and clear instructions	Thank-you screen with NP ICT branding

SITEMAP



SUPPORTING VISUALS





COMPREHENSIVE RESEARCH

1. Popularity of Photo Booth Experiences in Singapore

Photo booth experiences have become increasingly popular in Singapore, especially at events such as open houses, celebrations, and social gatherings. Well-known local examples include Solace Studios and Life4Cut, which are widely used for their visually appealing setups and easy-to-use photo experiences. These services attract users by offering themed designs, creative visuals, and memorable photo outputs that users enjoy sharing on social media.

The popularity of these photo booths shows that users value interactive experiences that go beyond traditional photography. This supports the idea of introducing a photo booth kiosk within the School of InfoComm Technology to engage students and visitors in a similar, familiar way.

2. User Engagement Through Visual and Interactive Elements

Services like Solace Studios and Life4Cut are successful because they focus on visual engagement. Users are encouraged to take multiple photos due to attractive backgrounds, lighting, and themed elements. This increased engagement aligns with experiential design principles, where interaction and enjoyment play a key role in shaping positive user experiences.

Building on this, the NP ICT AR Photo Booth Kiosk introduces augmented reality (AR) filters to enhance visual interest digitally rather than through physical props. AR overlays such as themed graphics or funny effects allow users to personalize their photos while keeping the kiosk compact and easy to maintain.

COMPREHENSIVE RESEARCH

3. Ease of Use and Accessibility

A huge reason behind the success of popular photo booths is their simplicity. Users can quickly understand how to use the booth without instructions or assistance. Most photo booth experiences follow a straightforward flow, which is, pose, take photo, review, and download.

The NP ICT AR Photo Booth Kiosk's direct user flow, which reduces steps and makes use of large buttons and clear visuals, was influenced by this finding. Designing for simplicity is especially important in public kiosks where users may be first-time visitors or unfamiliar with the system.

4. Photo Sharing and Digital Outputs

Another important feature observed in services like Life4Cut is the ease of accessing photo outputs. Users prefer fast and convenient methods such as QR code downloads or digital delivery, as these options reduce waiting time and remove the need for additional applications.

In response to this user behaviour, the proposed kiosk includes QR code download. This ensures that users can quickly save and share their photos, increasing satisfaction and encouraging online sharing that promotes NP ICT's identity.

COMPREHENSIVE RESEARCH

5. Justification for AR Integration

While traditional photo booths rely on physical props and backdrops, there is a growing trend toward digital enhancements such as AR filters, especially among younger users familiar with social media platforms. AR features provide flexibility, allow easy theme updates, and offer a more modern experience.

By integrating AR filters into the photo booth kiosk, the proposed design builds on the success of existing photo booth services while offering a more innovative and technology-driven experience that aligns with the identity of the School of InfoComm Technology.

6. Competitive Analysis

From a competitive perspective, existing photo booth services such as Solace Studios and Life4Cut are effective in engaging users through visual appeal and ease of use. However, they rely heavily on physical setups and lack digital flexibility.

The proposed AR photo booth kiosk addresses this gap by introducing customizable AR filters and a compact kiosk format, offering a more adaptable and technology-driven solution.

FIGMA LINK

<https://www.figma.com/design/TsWfalokRbgWmHSQ0es3iY/EXD-Asg2?node-id=0-1&t=e0KgXzjZk4spqLml-1>

BUDGET BREAKDOWN

1	Budget Plan			
2	Category	Quantity	Item	Estimated Cost (SGD)
3	Hardware	1	Touchscreen display	1800
4	Hardware	1	HD Camera	350
5	Hardware	1	LED Ring Light	200
6	Hardware	1 Set	Speakers	120
7	Hardware	1	Mini PC	900
8	Hardware	1	Thermal Photo Printer	700
9	Build	1	Custom Kiosk Structure	1500
10	Build	1 Set	Mounting & Acrylic Panels	350
11	Decals	1 Set	Stickers & Branding	200
12	Software	1	AR Filter & UI Design	500~
13	Misc	1 Set	Cables, Accessories & Buffer	530
14			Total Estimated Cost	7150

This budget plan shows a reasonable estimate of the key components required to build one interactive AR photo booth kiosk. All quantities are based on a single kiosk unit and remain within the \$8000 budget limit.

REFERENCES

Solace Studios: <https://www.solacestudios.sg/>

Life4Cut: <https://life4cutsg.net/>